BOUNDED BELOW COMPLEXES ARE DETECTED BY A SINGLE PERFECT GENERATOR

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file perfect.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Editorial note. One sentence inside the counted proof reads, verbatim: 'Small detail omitted.' (at the end of the paragraph extending the property from the generator to all perfect complexes, where the author says the same argument works for the remaining properties using that duality is a self equivalence). The counted span is otherwise the author's complete proof of the proposition; per the project rule the phrase is recorded here verbatim.

Allowed standard prerequisites (editorial).

- A map out of a perfect complex into a bounded below complex of quasi-coherent modules vanishes in low degrees, so condition (2) follows from boundedness below. invoked at proof lines 10859-10862. Gives the implication from (1) to (2).
- The Hom groups of the derived category, the Ext groups and the derived Hom, as well as the dual of a perfect complex and the global sections of the derived tensor product, are related by the standard identifications. used at proof lines 10863-10872. Shows that conditions (2) to (6) are all equivalent.
- A perfect complex generating the derived category is a classical generator, and a property of objects that is inherited by direct sums, summands and shifts and satisfies the two out of three property holds for all perfect complexes. invoked at proof lines 10875-10884. Extends the equivalent conditions from the given generator to every perfect object, giving condition (7).
- The induction principle on quasi-compact opens reduces a statement about all quasi-compact quasi-separated schemes to the affine case and a union step. invoked at proof lines 10887-10950. Reduces the converse to the affine case and the union of two opens.

Source reference. Original lemma 09M4: sources/stacks-perfect/pinned-original-source.tex, source lines 3465–3465. Invoked at proof lines 10859-10862 for the implication from (1) to (2); the shift argument that makes it applicable is inside the counted span.

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Source reference. Original section 0B6A: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10863-10866 for the equivalence of conditions (2), (3), (4); the dual identification that links them to (5), (6) is inside the counted span.

Source reference. Original lemma 08DQ: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 10869-10872 to see that the block (2) to (4) is equivalent to the block (5), (6); the observation about the dual of a shift is inside the counted span.

Source reference. Original remark 0GEF: sources/stacks-perfect/pinned-original-source.tex, source lines 4506–4506. Cited at proof lines 10874-10876 to apply the classical generator criterion; the verification that the property has the required inheritance properties is inside the counted span.

Source reference. Original remark 0ATH: sources/stacks-chow/tags.txt, source lines 1–1. Used at proof lines 10879-10883 to extend the property from the generator to all perfect complexes; the inheritance verification that makes it applicable is inside the counted span.

Source reference. Original lemma 08DR: sources/stacks-chow/tags.txt, source lines 1–1. Invoked at proof lines 10887-10889 to organise the converse; the affine case and the union step are inside the counted span.

Source reference. Original lemma 06Z0: sources/stacks-perfect/pinned-original-source.tex, source lines 454–454. Used at proof lines 10891-10894 to conclude boundedness below in the affine case; the choice of the structure sheaf as test object is inside the counted span.

Source reference. Original lemma 09IQ: sources/stacks-perfect/pinned-original-source.tex, source lines 4058–4058. Invoked at proof lines 10896-10897 together with the theorem on generators to produce the perfect complex Q generating on the open U; the use of Q in the union step is inside the counted span.

Source reference. Original theorem 09IS: sources/stacks-perfect/pinned-original-source.tex, source lines 4111–4111. Cited with the previous label at proof lines 10896-10897 for the existence of the generating perfect complex on the open; the subsequent Koszul construction is inside the counted span.

Source reference. Original lemma 08DF: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10910-10913 for the two descriptions of the auxiliary perfect complex K prime; the computation that K prime is supported on the complement is inside the counted span.

Source reference. Original lemma 08DP: sources/stacks-chow/tags.txt, source lines 1–1. Cited with the previous label at proof lines 10912-10914 to see that K prime is perfect; the use of its second description is inside the counted span.

Source reference. Original lemma 0GEK: sources/stacks-perfect/pinned-original-source.tex, source lines 10629–10629. Applied at proof lines 10920-10924 to obtain the integer a and the truncation isomorphism with the direct image from the intersection; producing the vanishing it needs is inside the counted span.

Source reference. Original lemma 0GEM: sources/stacks-perfect/pinned-original-source.tex, source lines 10695–10695. Used twice at proof lines 10925-10930 to

transfer the vanishing to the open U; the identification of the truncations that makes this possible is inside the counted span.

Source reference. Original section 0716: sources/stacks-chow/tags.txt, source lines 1–1. Cited at proof lines 10945-10948 to conclude that the truncation of E is bounded below on X; the final deduction is inside the counted span.

Proposition 0.2. *Let X be a quasi-compact and quasi-separated scheme. Let $G \in D_{perf}(\mathcal{O}_X)$ be a perfect complex which generates $D_{QCoh}(\mathcal{O}_X)$. Let $E \in D_{QCoh}(\mathcal{O}_X)$. The following are equivalent*

1. *$E \in D^+_{QCoh}(\mathcal{O}_X)$,*
2. *$\mathrm{Hom}_{D(\mathcal{O}_X)}(G[-i], E) = 0$ for $i \ll 0$,*
3. *$\mathrm{Ext}^i_X(G, E) = 0$ for $i \ll 0$,*
4. *$R\mathrm{Hom}_X(G, E)$ is in $D^+(\mathbf{Z})$,*
5. *$H^i(X, G^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} E) = 0$ for $i \ll 0$,*
6. *$R\Gamma(X, G^\vee \otimes^{\mathbf{L}}_{\mathcal{O}_X} E)$ is in $D^+(\mathbf{Z})$,*
7. *for every perfect object P of $D(\mathcal{O}_X)$*
   (a) *the assertions (2), (3), (4) hold with G replaced by P, and*
   (b) *$H^i(X, P \otimes^{\mathbf{L}}_{\mathcal{O}_X} E) = 0$ for $i \ll 0$,*
   (c) *$R\Gamma(X, P \otimes^{\mathbf{L}}_{\mathcal{O}_X} E)$ is in $D^+(\mathbf{Z})$.*

Proof. Assume (1). Since $\mathrm{Hom}_{D(\mathcal{O}_X)}(G[-i], E) = \mathrm{Hom}_{D(\mathcal{O}_X)}(G, E[i])$ we see that this is zero for $i \ll 0$ by Lemma 09M4. This proves that (1) implies (2).

Parts (2), (3), (4) are equivalent by the discussion in Cohomology, Section 0B6A. Part (5) and (6) are equivalent as $H^i(X, -) = H^i(R\Gamma(X, -))$ by definition. The equivalent conditions (2), (3), (4) are equivalent to the equivalent conditions (5), (6) by Cohomology, Lemma 08DQ and the fact that $(G[-i])^\vee = G^\vee[i]$.

It is clear that (7) implies (2). Conversely, let us prove that the equivalent conditions (2) – (6) imply (7). Recall that G is a classical generator for $D_{perf}(\mathcal{O}_X)$ by Remark 0GEF. For $P \in D_{perf}(\mathcal{O}_X)$ let $T(P)$ be the assertion that $R\mathrm{Hom}_X(P, E)$ is in $D^+(\mathbf{Z})$. Clearly, T is inherited by direct sums, satisfies the 2-out-of-three property for distinguished triangles, is inherited by direct summands, and is preserved by shifts. Hence by Derived Categories, Remark 0ATH we see that (4) implies T holds on all of $D_{perf}(\mathcal{O}_X)$. The same argument works for all other properties, except that for property (7)(b) and (7)(c) we also use that $P \mapsto P^\vee$ is a self equivalence of $D_{perf}(\mathcal{O}_X)$. Small detail omitted.

We will prove the equivalent conditions (2) – (7) imply (1) using the induction principle of Cohomology of Schemes, Lemma 08DR.

First, we prove (2) – (7) $\Rightarrow$ (1) if X is affine. Let $P = \mathcal{O}_X[0]$. From (7) we obtain $H^i(X, E) = 0$ for $i \ll 0$. Hence (1) follows since E is determined by $R\Gamma(X, E)$, see Lemma 06Z0.

Now assume $X = U \cup V$ with U a quasi-compact open of X and V an affine open, and assume the implication (2) – (7) $\Rightarrow$ (1) is known for the schemes U, V, and $U \cap V$. Suppose $E \in D_{QCoh}(\mathcal{O}_X)$ satisfies (2) – (7). By Lemma 09IQ and Theorem 09IS there exists a perfect complex Q on X such that $Q|_U$ generates $D_{QCoh}(\mathcal{O}_U)$. Let $f_1, \ldots, f_r \in \Gamma(V, \mathcal{O}_V)$ be such that $V \setminus U = V(f_1, \ldots, f_r)$ as subsets of V. Let

$K \in D_{perf}(\mathcal{O}_V)$ be the object corresponding to the Koszul complex on $f_1, \ldots, f_r$. Let $K' \in D_{perf}(\mathcal{O}_X)$ be

$$K' = R(V \to X)_* K = R(V \to X)_! K, \tag{0.2.1}$$

see Cohomology, Lemmas 08DF and 08DP. This is a perfect complex on X supported on the closed set $X \setminus U \subset V$ and isomorphic to K on V. By assumption, we know $R\mathrm{Hom}_{\mathcal{O}_X}(Q, E)$ and $R\mathrm{Hom}_{\mathcal{O}_X}(K', E)$ are bounded below.

By the second description of K' in (0.2.1) we have

$$\mathrm{Hom}_{D(\mathcal{O}_V)}(K[-i], E|_V) = \mathrm{Hom}_{D(\mathcal{O}_X)}(K'[-i], E) = 0$$

for $i \ll 0$. Therefore, we may apply Lemma 0GEK to $E|_V$ to obtain an integer a such that $\tau_{\leq a}(E|_V) = \tau_{\leq a} R(U \cap V \to V)_*(E|_{U \cap V})$. Then $\tau_{\leq a} E = \tau_{\leq a} R(U \to X)_*(E|_U)$ (check that the canonical map is an isomorphism after restricting to U and to V). Hence using Lemma 0GEM twice we see that

$$\mathrm{Hom}_{D(\mathcal{O}_U)}(Q|_U[-i], E|_U) = \mathrm{Hom}_{D(\mathcal{O}_X)}(Q[-i], R(U \to X)_*(E|_U)) = 0$$

for $i \ll 0$. Since the Proposition holds for U and the generator $Q|_U$, we have $E|_U \in D^+_{QCoh}(\mathcal{O}_U)$. But then since the functor $R(U \to X)_*$ preserves bounded below objects (see Cohomology, Section 0716) we get $\tau_{\leq a} E \in D^+_{QCoh}(\mathcal{O}_X)$. Thus $E \in D^+_{QCoh}(\mathcal{O}_X)$. $\square$